

Adjunct Report — Electric Current and the Sulfate Energy Cycle

A terrestrial existence proof, an energetic effect, and a precise list of what hasn't been tested

A short supplement to the Gate 5 reports. It extends the “What we did not explore” thread (see `UV_sulfur_hybrid_note.md`) with one specific, real, terrestrial body of evidence: that electric current and a sulfur energy cycle already go together in the lab. It is a direction, not a result — and the last section maps, at length, the basic Earth-bound research that would have to be done before anyone could honestly design a sulfur-oxide-powered organism.

How to read the tiers

This adjunct uses the project's standing convention. [LAB] is traceable to a published laboratory result, and where it is load-bearing the evidence is the primary paper, cited — not this document's restatement of it. [SPEC] is reasoned speculation with no empirical anchor. A tag travels with its claim; a [SPEC] is never quietly absorbed into a confident sentence. Per project source discipline, the two prior LLM-generated BMFC syntheses that prompted this note are not cited as evidence — they were a map; every load-bearing claim below was re-verified against the primary literature.

One discipline point that turns out to be the story: where the prompting material carried specific numeric optima (“+600 mV”, “−0.4 V”), the primary literature shows these are organism- and system-specific and span a wide range. The honest claim is that anode potential tunes the system, not that any single optimum is universal. That caution is stated as a finding, not a footnote.

1. The demonstrated baseline — what benthic microbial fuel cells already do [LAB]

The load-bearing fact is simple and real: on Earth, a sulfur cycle and an electric current already run together on an electrode, in the laboratory, today.

In a benthic/sediment microbial fuel cell, sulfate-reducing bacteria (SRB) use organic carbon to reduce sulfate to sulfide; sulfide-oxidizing bacteria (SOB) then oxidize that sulfide to elemental sulfur, passing the liberated electrons to an anode. Lee and colleagues built exactly this SRB + SOB co-biofilm and showed sulfate → sulfide → S⁰

proceeding with electron transfer to the electrode, with the rate of current set by how fast sulfide is shuttled cell-to-cell between the two populations [LAB — Lee et al., *Bioresour. Technol.*, 2014; 2012 precursor]. Independently, sulfide-oxidizing bacteria drawn from biodesulfurization reactors were shown to shuttle electrons from sulfide straight to an anode, producing measurable current density (~ 0.48 and ~ 0.24 A/m² for two communities, at +0.1 V vs Ag/AgCl) [LAB — de Rink et al., *Environ. Sci. Technol. Lett.*, 2018].

The biology can also wire a sulfur gradient over macroscopic distances. Cable bacteria — centimetre-long filamentous Desulfobulbaceae — couple sulfide oxidation in deep anoxic sediment to oxygen reduction at the surface by carrying electrons along their length, a result first shown by direct observation of the conducting filaments [LAB — Pfeffer et al., *Nature* 491:218–221, 2012] and grounded in the prior demonstration that electric currents couple spatially separated biogeochemical processes in marine sediment [LAB — Nielsen et al., *Nature* 463:1071–1074, 2010]. The underlying extracellular-electron-transfer (EET) machinery that makes any of this possible is the canonical one: *Geobacter sulfurreducens* respiring an electrode as its sole electron acceptor [LAB — Bond & Lovley, *Appl. Environ. Microbiol.* 69:1548–1555, 2003], and — directly relevant here — EET has now been characterized in a genuine *sulfate*-reducing bacterium, with both direct and indirect routes operating [LAB — Hou et al., 2024].

Two caveats belong in the baseline, not buried under it. First, in SRB-only anodes a large part of the current comes from the abiotic oxidation of biologically produced sulfide, not from a biological EET step — the biology makes the sulfide; chemistry often makes the electron [LAB — multiple BES reviews]. Second, elemental sulfur deposited at the anode tends to passivate the electrode, blocking further current; this is a recurring, real failure mode of sulfur bioelectrochemistry [LAB — sulfide-oxidation BES literature]. The existence proof is genuine, but it is a proof that the coupling can be made to work, not that it works cleanly or for free.

What §1 establishes: current and a sulfur cycle are demonstrably compatible on an electrode, with real organisms, in water — including a route to elemental sulfur and a biology that can transport electrons across centimetres. That is the floor the rest of this note stands on.

2. What current does to sulfate energetics — the driving-force argument [LAB → SPEC]

Why should current matter to a sulfur metabolism, beyond being a convenient output? Because sulfate reduction is a low-yield metabolism. It sits near the bottom of the redox tower: the free-energy change available per electron is small compared with, say, aerobic respiration or denitrification. This is classic, settled thermodynamics [LAB — Thauer, Jungermann & Decker, *Bacteriol. Rev.* 41:100–180, 1977], and the qualitative consequence — that organisms living this low on the tower grow slowly because so little energy is available — is independently visible across the comparative literature.

The relevant laboratory observation is that poisoning the anode potential changes how hard the biology works. Controlling the anode potential sets both the theoretical energy gain available to the microorganisms and the electrical output of the cell; in a continuous acetate-fed system, the reactor held at the favourable potential produced about 15% more charge and the highest peak power density of the set [LAB — Aelterman et al., *Appl. Microbiol. Biotechnol.* 78:409–418, 2008]. In a *sulfur-lineage* organism specifically — *Desulfuromonas acetexigens*, which reduces elemental sulfur — anode potential strongly influenced current and coulombic efficiency, with the best performance at -0.4 V vs Ag/AgCl and redox-dependent regulation of the EET pathways themselves [LAB — Sysoev, Katuri, Shaw & Saikaly, *Bioresour. Technol. Reports* 32:102448, 2025].

Two honest qualifications sit on this argument and must travel with it:

- **The optima do not generalize.** Across these studies the best operating potential ranged from roughly -400 mV to $+200$ mV vs Ag/AgCl depending on organism and system, and in the Aelterman work the three reactors converged to the same power density by the end of the run. “Applied potential tunes the metabolism” is [LAB]; “the optimum is X mV” is system-specific and not transferable.
- **The model organism is not doing the target metabolism.** *D. acetexigens* is a sulfur (S^0) reducer, not a sulfate reducer, and it is fed acetate, not run as a chemolithotrophic sulfur cycle. It demonstrates that a sulfur-lineage EET organism is potential-tunable — not that current has been shown to lift sulfate reduction toward viability.

With those guardrails, the bridge to the triboic idea is a clean [SPEC]: if a marginal sulfur metabolism sits just below the energy it needs, and applied potential demonstrably raises the thermodynamic driving force on related electrode-coupled metabolisms, then a current-supplemented sulfur metabolism is a thermodynamically coherent idea. A tribowhip is, physically, a current source. So “tribowhip current topping up a weak sulfur energy cycle” is conceptually coherent — exactly the supplement-to-a-weak-primary-cycle architecture the UV+sulfur note describes

— but the lift has been demonstrated for acetate and sulfur-reduction systems in water, never for the sulfur chemistry or the medium Venus actually presents.

What §2 establishes: the energetic logic (“current raises driving force; sulfate reduction is exactly the marginal metabolism that would benefit”) is real and grounded on Earth, and the extrapolation to a Venus sulfur metabolism is honest speculation gated on the chemistry §3 lists.

3. The Venus translation gaps — each broken Earth assumption [SPEC]

Every result in §§1–2 was obtained under conditions Venus does not provide. Each broken assumption is an explicit unmeasured-chemistry gap, and for each the lab work lives on Earth, not on a probe.

1. **Solvent.** Every BMFC above runs in water. Venus’ cloud droplets are concentrated sulfuric acid. EET cytochromes, the conductive fibres of cable bacteria, and biofilm matrices are all aqueous-evolved protein/polymer assemblies; none has been shown to conduct, fold, or survive in concentrated H_2SO_4 . *Unmeasured: does any electron-transfer-competent structure function in concentrated acid at all?*
2. **pH.** The demonstrations sit near neutral pH. Venus droplets are extraordinarily acidic. Redox potentials, sulfur speciation, and the stability of every intermediate shift with acidity; the Earth numbers do not transfer. *Unmeasured: the redox tower itself — the relative positions of the sulfur couples — at Venus acidity.*
3. **The terminal electron acceptor.** Earth BMFCs close the circuit with O_2 at the cathode (or nitrate). Venus has no free O_2 . A sulfur energy cycle there must close on some other acceptor (a sulfur oxidation state, CO_2 -derived species, or the triboelectric circuit itself). *Unmeasured: what plays the cathode’s role when there is no oxygen.*
4. **The sulfur inventory.** Earth systems run on dissolved sulfate and sulfide in solution. Venus’ sulfur is SO_2 , OCS , and S_8/S_x in a gas/aerosol system, with no large dissolved-sulfate reservoir — and, critically, the project already found (Gate 4, Exchange C) that $\text{SO}_2 \rightarrow \text{S}^0$ conversion is already explained abiotically by the Krasnopol’sky cycle. So the available sulfur is in the wrong chemical form and the obvious product is already accounted for without biology. *Unmeasured: whether any electrode-coupled metabolism can run on SO_2/OCS rather than dissolved sulfate, and whether it could leave a signature distinguishable from the abiotic baseline.*
5. **Electrode passivation, in acid.** The S^0 -passivation failure mode from §1 happens in water. In concentrated acid the deposition, dissolution, and morphology of elemental sulfur differ, and whether a biological “electrode” (a charge skin, a cell surface) would foul, dissolve, or survive is unknown. *Unmeasured: sulfur deposition behaviour at a bio-electrode in concentrated H_2SO_4 .*
6. **Water activity and the whole solvent role of water in metabolism.** Beyond being the solvent, water is a reactant and proton shuttle throughout sulfur biochemistry. At Venus water activity it is scarce. *Unmeasured: whether a sulfur energy cycle can be written that does not depend on abundant water as a reactant.*

What §3 establishes: the Earth existence proof breaks at six independent points on the way to Venus, and **every one of them is a wet-lab question that has not been asked**, not a modelling gap a laptop can close.

4. What it means for triboic life [SPEC]

Put the three pieces together. There is a real terrestrial existence proof that current and a sulfur cycle co-operate on an electrode (§1); a real energetic effect whereby applied potential raises the driving force on

exactly the kind of marginal metabolism sulfate reduction is (§2); and a triboelectric is a current source. So the original triboelectric intuition recorded in `uv_sulfur_hybrid_note.md` — electricity as a supplement to a weak primary sulfur energy cycle — is conceptually coherent and now has a concrete terrestrial analogue to point at, where before it had almost none.

That is the entire upgrade this note delivers, and it is worth being precise about its size. It moves the sulfur-supplement idea from “no benchmark at all” to “one real benchmark, in the wrong solvent, at the wrong pH, with the wrong acceptor, on the wrong sulfur species.” That is genuine progress and it is also, honestly, a long way from a result. It is the sulfur analogue of the project’s deepest carbon caveat (biosynthesis-in-acid, V-1): an idea whose physics is coherent but whose foundational chemistry is simply untested. It remains, exactly as the hybrid note said, a direction gated on laboratory work, not a result.

The natural question this raises — and the reason the supplement matters — is the one Report B’s bottom line already pointed at: the carbon creature could not explain Venus’ flaunted sulfur anomaly, and the shape of that failure points toward a sulfur-based organism the project deliberately did not build. If anyone were to build it, §5 is the bench-work that would have to come first.

5. The benchmark gap map — what basic Earth research must happen before a sulfur-oxide energy-cycle organism can be honestly designed

(On terminology: there isn’t a clean standing term for the metabolism this organism would run. “Sulfur-oxide-powered” is the most accurate plain description — an energy cycle whose external currency is the sulfur oxides SO_2 / OCS rather than dissolved sulfate. Below it is called the sulfur-oxide energy cycle [SPEC term].)

The point of this section is to be honest about why the project built the carbon creature and not this one: the carbon cycle had laboratory anchors at nearly every step, and the sulfur-oxide cycle has almost none. Below, the missing anchors are grouped from the most foundational (without which nothing else can be defined) to the most design-specific. Every item is Earth-bound, pre-Venus, and almost all are

wet-lab or bench-electrochemistry — none is a computation the project could have run. Each is marked with the discipline it falls in.

Tier I — Foundational chemistry (without these, no metabolism can even be written)

1. **Sulfur redox thermodynamics in concentrated H_2SO_4 .** (*physical/analytical chemistry*) The entire redox tower — the relative free energies of SO_2 , OCS, S_8/S_x , sulfite, thiosulfate, sulfate couples — needs to be measured at Venus acidity and water activity, not extrapolated from aqueous tables. Until the tower is known in this medium, “marginal metabolism that current could lift” has no quantitative meaning, because there is no measured ΔG to lift. *This is the keystone; almost everything below depends on it.*
2. **Speciation and kinetics of the sulfur oxides in concentrated acid.** (*physical chemistry*) What SO_2 , OCS, and elemental sulfur actually are in concentrated H_2SO_4 — dissolved, complexed, protonated, polymerized — and how fast they interconvert. The abiotic Krasnopolsky network is defined for the gas phase; the condensed-acid-phase chemistry that a droplet-dwelling organism would meet is far less characterized.
3. **Stability and turnover of sulfur intermediates in acid.** (*physical chemistry*) Which intermediates persist long enough to be metabolically useful versus which are destroyed or abiotically short-circuited. A metabolism cannot be built on an intermediate the medium consumes faster than an enzyme could.
4. **Water as reactant, not just solvent.** (*physical/biochemistry*) Whether any sulfur energy cycle can be written that does not require abundant water as a proton shuttle and reactant — and if water is required, what minimum water activity makes it run. This gates whether a low-water-activity organism is even thermodynamically allowed to use this chemistry.

Tier II — Electron-transfer machinery in a hostile medium (the EET problem)

5. **Does any electron-transfer-competent structure work in concentrated H_2SO_4 ?** (*biochemistry / bioelectrochemistry*) Every demonstrated EET conduit — c-type cytochromes, the cable-bacteria fibre, multiheme proteins — is aqueous-evolved. The single most important experiment is whether any conductive biological or biomimetic structure retains conductivity and integrity in concentrated acid. A negative result would be as decisive as the carbon creature’s biosynthesis-in-acid wall.
6. **Acid-stable charge-transfer chemistry (biomimetic allowed).** (*synthetic / materials chemistry*) If natural cytochromes fail, is there a synthetic or hybrid charge-transfer moiety — the project’s redox-core work explored quinone/semiquinone and ferrocene couples for a related reason — that can shuttle electrons at the needed potentials in this medium? This connects directly to the existing redox-core experiments and is the most tractable bench entry point.
7. **Electrode/bio-electrode behaviour and sulfur passivation in acid.** (*electrochemistry*) The S^0 -passivation failure mode must be characterized in concentrated acid: does elemental sulfur deposit, dissolve, or re-speciate at a charge surface here, and can a metabolism avoid fouling itself? This is a direct, runnable bench-electrochemistry experiment.
8. **Coupling current to a sulfur redox step — the core demonstration.** (*bioelectrochemistry*) The keystone biological experiment: show, even in water first and then toward acid, that an applied current measurably lifts the rate or yield of a sulfur-oxide (SO_2/OCS) transformation — i.e. reproduce the §2 driving-force effect on the target chemistry rather than on acetate or S^0 reduction. Nothing in the current literature does this for sulfur oxides.

Tier III — Closing the energy cycle without oxygen

9. **An O_2 -free terminal acceptor.** (*microbiology / electrochemistry*) Every BMFC closes on oxygen or nitrate. The sulfur-oxide organism needs a demonstrated acceptor that exists on Venus — another sulfur oxidation state, a CO_2 -derived species, or the triboelectric circuit itself acting as the sink. Which acceptors can close a sulfur energy loop anaerobically, and at what yield, is unmeasured.
10. **Whole-cycle energy balance with the supplement included.** (*bioenergetics*) Once 1, 8, and 9 exist, the actual experiment the whole idea rests on: measure whether primary sulfur turnover plus a current

supplement nets positive over maintenance — the sulfur analogue of the project's Gate 3b ledger, which for the carbon creature closed only on a knife-edge. Without the measured tower (item 1) this ledger cannot even be written.

11. **Maintenance and repair cost in concentrated acid.** (*biochemistry / materials*) The acid degrades structure continuously; the cycle must out-earn that degradation. The repair rate versus acid-degradation rate is the term that flipped the carbon creature's ledger, and it is entirely unmeasured for any sulfur system in this medium.

Tier IV — Distinguishability and organism-level design

12. **A biotic-vs-abiotic discriminator for the products.** (*isotope / analytical geochemistry*) Gate 4 already showed the obvious product (S^0 from SO_2) is abiotically explained. So before designing the organism, one must establish — in the lab — what isotopic ($\delta^{34}S$, $\Delta^{33}S$) or morphological signature a current-supplemented sulfur metabolism would leave that the Krasnopolsky cycle does not. Without this, even a working organism would be undetectable and the design uninformative.
13. **Light-element fractionation under current supplementation.** (*isotope geochemistry*) Whether driving a metabolism with external current changes its isotopic fractionation relative to an unforced one — a basic measurement with no current answer, and one that determines whether the headline discriminator even survives the supplement.
14. **Population/throughput limits in a droplet medium.** (*microbial ecology / transport*) The cell-to-cell sulfide-shuttling distance set the current in Lee's co-biofilm; the analogous transport limits (how dense, how fast, how far apart) in a sulfuric-acid aerosol droplet are unmeasured and bound any organism-scale design.

The honest bottom line of §5. The carbon creature was buildable on a laptop because items analogous to 1–14 mostly already had laboratory answers in the carbon literature. For the sulfur-oxide energy cycle, **item 1 alone — the redox tower in concentrated acid — is unmeasured, and it is the keystone the other thirteen hang from.** This is why the project built the tractable version and recorded this one as a direction. The work that would change that is not more modelling; it is roughly a dozen distinct wet-lab and bench-electrochemistry programs, beginning with measuring sulfur redox thermodynamics in concentrated sulfuric acid. Until those benchmarks exist, designing a sulfur-oxide-powered organism would be building on unmeasured chemistry — the precise failure the project's source discipline exists to prevent.

Sources (re-verified against primary literature)

1. Aelterman, P., Freguia, S., Keller, J., Verstraete, W. & Rabaey, K. (2008). The anode potential regulates bacterial activity in microbial fuel cells. *Appl. Microbiol. Biotechnol.* 78(3):409–418. doi:10.1007/s00253-007-1327-8
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3. Lee, D.-J. et al. (2014). Sulfate and organic carbon removal by microbial fuel cell with sulfate-reducing bacteria and sulfide-oxidising bacteria anodic biofilm. *Bioresour. Technol.* (2012 precursor study).
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6. Nielsen, L. P. et al. (2010). Electric currents couple spatially separated biogeochemical processes in marine sediment. *Nature* 463:1071–1074.
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9. Hou, Y. et al. (2024). Co-occurrence of direct and indirect extracellular electron transfer mechanisms during electroactive respiration in a dissimilatory sulfate-reducing bacterium. PMC11705803.

Dutta/Rabaey 2010 (Water Res.) and Jørgensen 1982 (Nature) from the prompting material were not load-bearing and were not separately re-verified; they are omitted rather than cited unverified, per source discipline.

Tier summary: §1 [LAB] with two stated caveats; §2 [LAB] effect → [SPEC] Venus bridge, with numeric optima explicitly non-generalized; §3 [SPEC], six unmeasured-chemistry gaps; §4 [SPEC], a coherent untested direction; §5 [SPEC], a benchmark map of ~14 Earth-bound experiments gating any honest design. This is a direction gated on laboratory work, not a result.